

Université Catholique d'Afrique Centrale (UCAC)

Institut Catholique de Yaoundé (ICY)

COURSE TITLE: MATHEMATICAL ANALYSIS

DEPARTMENTS: BEM1

LECTURER: ARREY Marinus

INSTRUCTIONS: ANSWER ALL QUESTIONS

SECOND SEMESTER EXAMINATION

Question: (2marks)

Solve for X the following functions:

a) $\log_3(X+2) + \log_3(X+4) = 1$

b) $2^{2x} - 10(2^x) + 16 = 0$

Question: (4marks)

Given the demand function $2P = 60 - Q$, and the supply function $P = 12 + 4Q$, and assuming pure competition, calculate the consumer's surplus and the producer's surplus.

Question: (4marks)

The production function of a commodity is: $Q = 10L - 0.1L^2 + 15K - 0.2K^2 + 2KL$ where L is labour, K is capital and Q is production.

- Calculate the marginal products of the two inputs when 10 units of each labour and capital are used,
- If 10 units of capital are used, what is the upper limit for the use of labour which a rational producer will never exceeds.
- Determine the marginal rate of technical substitution (MRTS)

Question: (4marks)

If Q_1 and P_1 are the demand and prices of tea, Q_2 and P_2 are demand and price of coffee and demand function are given by: $Q_1 = P_1^{-1.7} P_2^{0.6}$; $Q_2 = P_1^{0.4} P_2^{-0.8}$. Calculate the two cross elasticities of demand and point out whether the commodities are competitive or complementary.

Question (6marks)

- What output mix should a profit-maximizing firm produce when its total profit function is $80X - 2X^2 - XY - 3Y^2 + 100Y$ and its maximum output capacity is $X+Y + 12$
- Estimate the effect on profits if output capacity is expanded by 1 unit.
- Verify whether it is a minima or a maxima